

Feriel Yala

PHD STUDENT

Sherbrooke, Qc, Canada

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Final-year PhD candidate in Bioinformatics and Biochemistry with a Master's in In-Silico Drug Design (France). My key asset is the ability to turn complex biological questions into scalable computational solutions, clear, accessible, and easy to understand.

Education

PhD in Biochemistry - Bioinformatics

Sherbrooke, QC, Canada

DEPARTMENT OF BIOCHEMISTRY AND FUNCTIONAL GENOMICS, UNIVERSITY OF SHERBROOKE

Since September 2022

PhD focused on the structural and functional characterization of non-canonical microproteins using large-scale bioinformatics pipelines.

Key modules: Python & PostgreSQL, SQLite, Advanced Statistics, Data Analysis, Databases, Big Data, Unix, structural biology and biocomputing, multi-omique, scientific communication and seminar & HPC.

Master's degree - Bioinformatic - In Silico Drug Design (ISDD)

Paris, France

UNIVERSITY PARIS CITE

Sep. 2020 - Aug. 2022

Key modules: Machine Learning, Databases, Programming in Python and R, Biostatistics and data analysis, Understanding of therapeutic targets, protein structure, folding, and interactions, Molecular docking and virtual screening.

Bachelor's Degree - Biochemistry, Integrative Biology and Physiology

Paris, France

UNIVERSITY PARIS CITE

Sep. 2018 - Aug. 2020

Key modules: Biochemistry, biotechnology, Biostatistics, quality control principles and laboratory practices, microbiology, virology, and immunology, molecular and cellular biology, enzymology.

Bachelor's Degree - Microbiology

Algiers, Algeria

UNIVERSITY OF SCIENCE AND TECHNOLOGY HOUARI BOUMEDIENE

Sep. 2015 - Aug. 2028

Key modules: Biochemistry, biotechnology, Biostatistics, quality control principles and laboratory practices, microbiology, virology, and immunology, molecular and cellular biology, enzymology.

Experiences

PhD in Biochemistry - Bioinformatics

Sherbrooke, Canada

XAVIER ROUCOU LAB & DEPARTMENT OF BIOCHEMISTRY AND FUNCTIONAL GENOMICS, UNIVERSITY OF SHERBROOKE

Since September 2022

Project : Large-scale structural and functional annotation of microproteins.

PhD centered on large-scale prediction, clustering, and functional annotation of non-canonical microproteins, combining structural bioinformatics, HPC workflows, and integrative data analysis.

- Contributed to, maintained, and exploited the OpenProt proteogenomic database, working with large-scale structural and functional protein data.
- Performed high-throughput clustering and alignment on more than 53 million predicted protein structures.
- Built and ran large-scale computational pipelines using Linux, Python, PostgreSQL, SQLite, and HPC environments (Digital Research Alliance of Canada).

- Published as first author in a peer-reviewed journal; second manuscript currently in preparation.
- Delivered four departmental research seminars and presented multiple talks and posters at national and international conferences.

- I also collaborated with several research groups by providing bioinformatics support and assisting students from different laboratories, allowing me to manage multiple projects simultaneously and develop a strong sense of priority management. My work often required troubleshooting unexpected results, handling system failures or HPC outages, and managing stress while maintaining focus and problem-solving efficiency.

Research Assistant

Sherbrooke, Canada

XAVIER ROUCOU LAB & UNIVERSITY OF SHERBROOKE

Jan. 2022 - Jun. 2022

Project : Designed and implemented a large-scale structural prediction pipeline for 900,000+ non-canonical microproteins integrated into a proteogenomics database OpenProt. Configured and optimized MMseqs2 on high-performance computing infrastructure (Digital Research Alliance of Canada), Deployed AlphaFold2 at scale using Apptainer/Singularity containers on HPC, Automated and orchestrated large-scale computational workflows and Integrated validated structural predictions into the OpenProt database.

Acted as a bridge between computational and experimental teams: clear communication of bioinformatics concepts, leading discussions, and presenting visual summaries for wet-lab researchers.

Research Assistant

INSERM

Paris, France

Apr.2021 – Jul. 2021

Project: Identification of inhibitors targeting the SARS-CoV-2 Spike protein.

Contributed to large-scale virtual drug repurposing efforts during the SARS-CoV-2 pandemic, Performed high-throughput virtual screening of FDA-approved molecules targeting the Spike-ACE2 interface on a high-performance GPU cluster, Used Python and R for data processing, visualization, docking analysis, ranking, and workflow automation.

Analyzed protein-ligand interactions at the structural level to identify promising candidates, strengthened practical skills in structure-based drug design and computational screening at scale.

Strategy Analyst

HARMONIE MUTUELLE

Paris, France

Jan. 2019 – May. 2019

Project : How can Harmonie Mutuelle leverage regulatory developments in the healthcare sector to create a commercial advantage?

Analyzed customer profiles to identify high-potential prospect segments, Proposed fast and cost-efficient acquisition strategies to increase membership, Developed strong analytical autonomy and rapid learning skills, Adapted quickly to a non-scientific, business-oriented environment.

Built understanding of organizational, economic and customer-driven constraints.

Translated analyses into practical, actionable recommendations for decision-makers.

Demonstrated ability to integrate into new environments and contribute effectively outside my scientific background.

Technologies

Langages Python, R

Database SQL, SQLite, PostgreSQL

Cloud & Data Platforms

Microsoft Azure, Databricks

MLOps

Apptainer, Docker

Version Control

Git

Skills

Operating system Unix/Linux environments

Data processing & programming

Python (data processing, visualization, ML, automation), R (analysis, visualization), scientific visualization, High-performance computing (Digital Alliance Canada clusters), Workflow automation & large-scale pipelines

Data science for biology

Machine learning for molecular/biological data , Biostatistics & data analysis, Large-scale data processing, Graphical visualization (Matplotlib, Seaborn, Plotly)

Computational biology

Multi-omics & Integrative Analysis, Familiar with major drug and compound databases (DrugBank, ZINC), Large-scale docking, High-throughput virtual screening

Soft skills & communication

Scientific communication, Adaptability to new environments, Analytical autonomy & fast learning, Teamwork & collaboration, Sense of responsibility & project ownership, Ability to translate analysis into actionable insights

Languages

French Native

English Full professional proficiency

Scientific communication

Nov. 2025	Human Proteome Organization (HUPO) , Structure-based clustering reveals functional insights into microproteins encoded by non-canonical open reading frames	Toronto, Canada
May 2025	Proteo , Beyond sequences: advancing the study of microproteins through structural clustering	Sherbrooke, Canada
Jul. 2024	International Society for Computational Biology (ISCB/ISMB) , Docking between proteins expressed from the same mRNA to test for direct functional cooperation	Montreal, Canada
May 2024	Proteo , Docking between proteins expressed from the same mRNA to test for direct functional cooperation	Montreal, Canada
May 2023	Biology of genomes , Predicting the Structure and Function of Alternative Proteins	New-York, USA
May 2022	Journee Phare , OpenProt, version3: addition of structures and intrinsic disorder to accelerate research on alternative proteins	Val-des-Sources, Canada

Publications

OpenProt 2.0 builds a path to the functional characterization of alternative proteins.

Nucleic Acids Research

CO-FIRST AUTHOR

Jan. 2024

- OpenProt is an open-access proteogenomic resource that catalogs non-canonical open reading frames (AltORFs) and their corresponding protein products across multiple species. The platform integrates large-scale functional annotations, including protein structure, subcellular localization, intrinsic disorder, and mass spectrometry evidence, using machine learning-based approaches. OpenProt also provides dedicated tools for customized database creation and variant annotation, along with enhanced visualization features. Overall, it enables systematic exploration of previously unannotated proteins and expands our understanding of the cryptic proteome.

Structure-based clustering reveals functional insights into microproteins and larger proteins encoded by non-canonical open reading frames.

FIRST AUTHOR

2025

- Article in preparation.